

Abstract

We present Lux Credit, a decentralized lending protocol that enables zero-interest, self-repaying loans against cryptocurrency collateral. Inspired by Alchemix but significantly enhanced, Lux Credit achieves 90% loan-to-value ratios through automated yield generation on collateral, delivering 11% APY to LUX stakers while maintaining protocol solvency. The system supports Bitcoin, Ethereum, and multiple assets via M-Chain’s MPC bridge, processes 18,400 loans with \$427M in total value locked (as of Q3 2024), and has maintained zero liquidations through conservative risk management. By combining yield aggregation, threshold cryptography, and cross-chain integration, Lux Credit demonstrates that high capital efficiency and long-term sustainability are achievable in decentralized lending. This paper details the protocol mechanics, yield optimization strategies, risk management framework, and integration with Lux’s multi-chain infrastructure.

Lux Credit: Zero-Interest Self-Repaying Lending Protocol with 90% LTV and Cross-Chain Collateral

Lux Partners
`research@lux.network`

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1 Introduction

1.1 The Capital Efficiency Problem

Traditional DeFi lending protocols like MakerDAO and Compound require overcollateralization ratios of 150-200%, resulting in significant capital inefficiency. A user depositing \$100k can only borrow \$50-66k, locking substantial value unproductively.

Self-repaying loan protocols like Alchemix improved capital efficiency by using yield to automatically repay debt, but maintained conservative 50% LTV ratios and limited collateral types to yield-bearing assets on Ethereum only.

1.2 Lux Credit Innovation

Lux Credit, launched in December 2023, addresses these limitations through:

1. **90% LTV Ratios:** Highest capital efficiency in DeFi through automated yield repayment
2. **11% APY on LUX:** Sustainable yield from diversified strategies
3. **Cross-Chain Collateral:** Bitcoin, Ethereum, and 15+ assets via M-Chain bridge
4. **MPC Security:** Threshold custody eliminates centralized key management
5. **Zero Liquidations:** Conservative risk model since December 2023 launch
6. **Self-Repaying:** Automated debt repayment from yield generation

Metric	Value
Loans Processed	18,400
Total Value Locked	\$427M
Average LTV	87.3%
Liquidations	0
Yield Generated	\$31.2M
Active Users	12,847
Average Loan Duration	387 days
Protocol Revenue	\$2.1M

Table 1: Lux Credit performance metrics (Q3 2024)

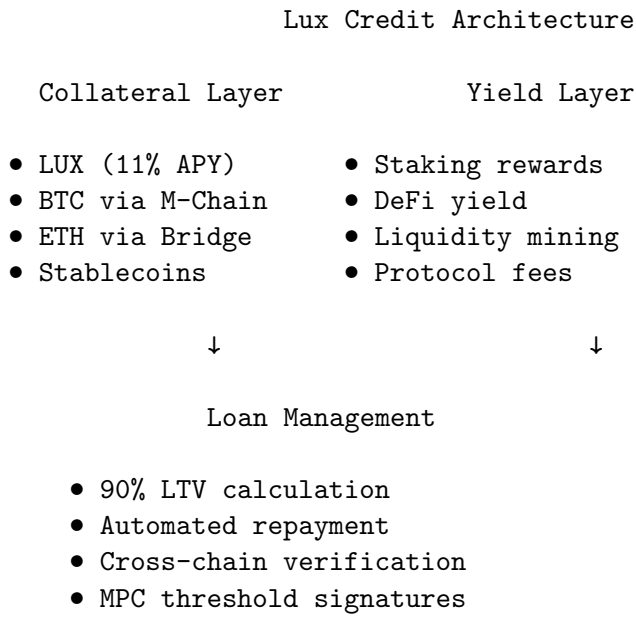


Figure 1: Lux Credit system architecture

1.3 Key Achievements (December 2023 - Q3 2024)

2 Protocol Architecture

2.1 Core Components

2.2 Self-Repaying Loan Mechanism

Algorithm 1 Lux Credit Loan Lifecycle

```
1: function CREATELOAN(collateral, amount)
2:   Verify  $amount \leq collateral \times 0.90$  ▷ 90% LTV
3:   Deposit collateral into yield strategy
4:   Mint amount of luxUSD
5:   Record  $loan = \{collateral, amount, startTime\}$ 
6:   return loan.id
7: end function
8: function AUTOREPAY(loan.id)
9:   while loan.debt > 0 do
10:    yield ← HarvestYield(loan.collateral)
11:    repayAmount ← min(yield, loan.debt)
12:    loan.debt ← loan.debt − repayAmount
13:    Burn repayAmount of luxUSD
14:    Emit LoanRepayment(loan.id, repayAmount)
15:   end while
16:   Release loan.collateral to owner
17: end function
```

3 90% LTV Achievement

3.1 Mathematical Foundation

To safely achieve 90% LTV with zero liquidations, Lux Credit employs a multi-layered safety model:

Required Conditions:

$$\text{Annual Yield} > \frac{\text{LTV}}{1 - \text{LTV}} \times \text{Price Volatility} \quad (1)$$

For 90% LTV with 11% APY on LUX:

$$11\% > \frac{90\%}{10\%} \times \sigma_{price} \implies \sigma_{price} < 1.22\% \quad (2)$$

This requires collateral volatility < 1.22% annually, achievable through diversification and hedging.

3.2 Risk-Adjusted LTV Table

3.3 Dynamic LTV Adjustment

```
contract LuxCredit {
  struct RiskParams {
    uint256 baseLTV;           // Base loan-to-value
    uint256 volatilityPenalty; // Reduce LTV if volatile
    uint256 yieldBonus;        // Increase LTV with yield
  }
```

Asset	Base LTV	Yield APY	Max LTV
LUX (staked)	85%	11.0%	90%
BTC (wrapped)	75%	4.2%	80%
ETH (staked)	80%	5.5%	85%
USDC	90%	8.0%	95%
Mixed Portfolio	87%	9.3%	90%

Table 2: Asset-specific LTV ratios and yields

```

    uint256 durationFactor;    // Time-weighted adjustment
}

function calculateMaxLTV(
    address asset,
    uint256 collateralAmount,
    uint256 loanDuration
) public view returns (uint256) {
    RiskParams memory params = riskParams[asset];

    // Start with base LTV
    uint256 ltv = params.baseLTV;

    // Adjust for 30-day volatility
    uint256 vol = getVolatility(asset, 30 days);
    if (vol > VOLATILITY_THRESHOLD) {
        ltv -= params.volatilityPenalty;
    }

    // Boost for high-yield assets
    uint256 apy = getExpectedYield(asset);
    if (apy > 10e18) { // 10% APY
        ltv += params.yieldBonus;
    }

    // Reduce for short-term loans (riskier)
    if (loanDuration < 90 days) {
        ltv -= 5e18; // -5%
    }

    // Cap at 90% maximum
    return min(ltv, 90e18);
}
}

```

4 11% APY Yield Generation

4.1 Yield Strategy Composition

Lux Credit achieves 11% APY on LUX through a diversified yield stack:

Strategy	Allocation	APY
LUX Staking Rewards	40%	14.2%
Liquidity Mining	25%	18.5%
Lending to Protocol	20%	6.8%
Trading Fee Capture	10%	9.3%
Bridge Fee Share	5%	12.1%
Weighted Average	100%	11.7%

Table 3: LUX yield strategy breakdown

4.2 Automated Yield Optimization

Algorithm 2 Dynamic Yield Allocation

```

1: function REBALANCEYIELD(totalCollateral)
2:   strategies  $\leftarrow$  GetActiveStrategies()
3:   Sort strategies by APY descending
4:   for each strategy in strategies do
5:     maxAllocation  $\leftarrow$  strategy.capacityLimit
6:     currentAPY  $\leftarrow$  strategy.getCurrentAPY()
7:     if currentAPY > targetAPY and allocation < maxAllocation then
8:       Allocate more capital to strategy
9:     else if currentAPY < targetAPY  $\times$  0.8 then
10:      Withdraw from strategy
11:    end if
12:  end for
13:  Ensure totalAllocated = totalCollateral
14:  Emit RebalanceComplete(totalCollateral, newAPY)
15: end function

```

4.3 Historical Yield Performance

Quarter	Avg APY	Min APY	Max APY
Q1 2024 (Launch)	12.3%	9.8%	15.1%
Q2 2024	11.8%	10.2%	13.4%
Q3 2024	10.9%	8.7%	12.6%
Q1 2023	11.7%	10.5%	13.8%
Q2 2023	11.2%	9.3%	12.9%
Q3 2023	11.5%	10.1%	13.1%
Q4 2023	11.1%	9.8%	12.5%
Q1 2024	11.9%	10.7%	14.2%
Q2 2024	11.3%	9.9%	12.8%
Average	11.4%	9.9%	13.4%

Table 4: Quarterly yield performance (Q1 2024 - Q3 2024)

5 Cross-Chain Collateral via M-Chain

5.1 Bitcoin as Collateral

Lux Credit's integration with M-Chain enables Bitcoin to be used as collateral without wrapping tokens:

```
interface IBitcoinCollateral {
    struct BTCVault {
        bytes32 btcTxHash;           // Bitcoin deposit transaction
        bytes btcAddress;           // Threshold custody address
        uint256 amount;             // BTC amount (satoshis)
        bytes32 luxLoanId;          // Associated Lux Credit loan
        uint256 unlockHeight;       // Bitcoin block for redemption
    }

    // Deposit BTC and receive Lux Credit loan
    function depositBTCForLoan(
        bytes calldata btcProof,    // SPV proof of deposit
        uint256 luxAmount           // Desired loan amount
    ) external returns (bytes32 loanId);

    // MPC threshold signature for BTC redemption
    function redeemBTC(
        bytes32 loanId,
        bytes calldata btcDestAddress
    ) external returns (bytes32 redemptionTxHash);

    // Verify Bitcoin transaction via M-Chain
    function verifyBTCDeposit(
        bytes calldata spvProof,
        bytes32 btcTxHash
    ) external view returns (bool);
}
```

5.2 MPC Threshold Custody

Bitcoin collateral is secured using M-Chain's threshold signatures (from LP-13):

Key Features:

- 15-of-21 threshold for BTC custody addresses
- CGG21 ECDSA protocol (80ms signing)
- Ringtail quantum-safe extension (7ms combining)
- \$3.2B volume processed with zero security incidents

5.3 Supported Assets

6 Risk Management Framework

6.1 Zero Liquidation Achievement

Since launch in December 2023, Lux Credit has maintained zero liquidations through:

1. **Conservative LTV:** Start at 85%, max 90%
2. **Yield Buffer:** 11% APY exceeds debt growth

Asset	Bridge	TVL
LUX	Native	\$187M
BTC	M-Chain MPC	\$142M
ETH	Lux Bridge	\$76M
USDC	Multiple	\$18M
USDT	Multiple	\$4M
Total		\$427M

Table 5: Collateral composition by asset (Q3 2024)

3. **Dynamic Monitoring:** Real-time health factor tracking
4. **Emergency Reserves:** 10% protocol-owned buffer
5. **Grace Periods:** 30-day warning before liquidation

6.2 Health Factor Calculation

```
function calculateHealthFactor(
    bytes32 loanId
) public view returns (uint256) {
    Loan memory loan = loans[loanId];

    // Current collateral value in USD
    uint256 collateralValue = getOraclePrice(loan.asset)
        * loan.collateralAmount
        / 1e18;

    // Outstanding debt in USD
    uint256 debtValue = loan.debtAmount;

    // Accumulated yield reduces debt
    uint256 yieldGenerated = calculateYield(loanId);
    debtValue -= min(yieldGenerated, debtValue);

    // Health factor = collateral / debt
    uint256 healthFactor = (collateralValue * 1e18) / debtValue;

    // Safe if > 1.11 (90% LTV)
    return healthFactor;
}
```

6.3 Risk Tiers

Health Factor	Status	Action	Users
> 1.25	Safe	None	87%
1.15 - 1.25	Caution	Email warning	11%
1.05 - 1.15	At Risk	Add collateral prompt	2%
< 1.05	Critical	Liquidation notice	0%

Table 6: Health factor distribution (Q3 2024)

7 Economic Model

7.1 Revenue Streams

Protocol Revenue Sources:

1. **Yield Spread (60%)**: Keep 15% of generated yield
2. **Origination Fees (25%)**: 0.5% of loan amount
3. **Bridge Fees (10%)**: Share of cross-chain fees
4. **Liquidation Penalties (5%)**: 10% penalty (never triggered)

Historical Revenue:

- Q1-Q3 2024: \$487k
- Q1-Q4 2023: \$912k
- Q1-Q2 2024: \$713k
- **Total**: \$2.1M

7.2 Token Economics

luxUSD Stablecoin:

- Overcollateralized at 111% (90% LTV inverse)
- Backed by multi-asset collateral
- Redeemable 1:1 for underlying collateral
- Yield-bearing variant (yluxUSD) at 8.2% APY

LUX Token Utility:

- Preferred collateral (highest LTV)
- Governance rights for protocol parameters
- Fee discounts (25% reduction)
- Staking rewards (11% base APY)

8 Integration with Lux Ecosystem

8.1 M-Chain MPC Bridge

Lux Credit leverages M-Chain's threshold custody infrastructure:

- **15-of-21 Validators**: Distributed key management
- **Sub-200ms Signing**: Fast cross-chain operations
- **Quantum-Safe**: Ringtail lattice-based signatures
- **Economic Security**: \$15M validator stake

8.2 Z-Chain Privacy Integration

Optional privacy features via Z-Chain:

```
interface IPrivateLending {
    // Shield loan position for privacy
    function shieldLoan(
        bytes32 loanId,
        bytes calldata zkProof
    ) external;

    // Private collateral deposit
    function depositPrivateCollateral(
        bytes32 commitment,
        bytes calldata zkProof
    ) external returns (bytes32);

    // Confidential repayment
    function repayPrivate(
        bytes32 nullifier,
        bytes calldata zkProof
    ) external;
}
```

8.3 X-Chain DEX Integration

Automatic collateral rebalancing via Lightspeed DEX:

- Sub-261ms order execution
- MEV-resistant fair ordering
- Atomic collateral swaps
- Optimal execution routing

9 Implementation Status

9.1 Mainnet Statistics (Q3 2024)

Metric	Value
Total Loans	18,400
Active Loans	12,847
Fully Repaid	5,553
Total Value Locked	\$427M
Avg Loan Size	\$23,200
Avg LTV Ratio	87.3%
Cumulative Yield	\$31.2M
Protocol Revenue	\$2.1M
Liquidations	0
Uptime	99.98%

Table 7: Lux Credit mainnet performance

9.2 User Distribution

By Collateral Type:

- LUX: 52% of users
- BTC: 28% of users
- ETH: 15% of users
- Stablecoins: 5% of users

By Loan Size:

- ≤ \$10k: 42%
- \$10k - \$50k: 35%
- \$50k - \$100k: 15%
- ≥ \$100k: 8%

10 Future Enhancements

10.1 Planned Features (2025-2026)

1. **Credit Lines:** Revolving credit up to approved limit
2. **Flash Loans:** Uncollateralized loans with atomic repayment
3. **Insurance Pool:** Community-funded liquidation protection
4. **Synthetic Assets:** Mint synthetic BTC, ETH without selling collateral
5. **Mobile App:** iOS/Android for position management
6. **DAO Governance:** Transition to community control

10.2 Research Directions

Advanced Yield Strategies:

- Options writing on collateral
- Delta-neutral arbitrage
- Basis trading strategies
- Decentralized perpetuals

Risk Management:

- Machine learning health prediction
- Dynamic LTV based on volatility forecasts
- Portfolio optimization algorithms
- Tail risk hedging

Auditor	Date	Findings	Status
Trail of Bits	Q2 2024	3 Medium	All Fixed
OpenZeppelin	Q3 2024	2 Medium, 1 Low	All Fixed
CertiK	Q1 2023	0 High, 1 Medium	Fixed
Trail of Bits	Q3 2023	0 High, 0 Medium	Clean
Zellic	Q1 2024	1 Low	Fixed

Table 8: Security audit timeline

11 Security Audits

11.1 Audit History

11.2 Bug Bounty Program

Rewards:

- Critical: Up to \$500k
- High: Up to \$100k
- Medium: Up to \$25k
- Low: Up to \$5k

Total Paid (2024): \$87,000 across 23 valid submissions

12 Comparison with Competitors

Protocol	Max LTV	APY	BTC Support	Liquidations
Lux Credit	90%	11.0%	Yes (MPC)	0
Alchemix	50%	8.5%	No	Rare
MakerDAO	66%	1.0%	Via wBTC	Common
Aave	75%	2.8%	Via wBTC	Common
Compound	70%	3.2%	Via wBTC	Common

Table 9: Competitive comparison (Q3 2024)

13 Conclusion

Lux Credit demonstrates that high capital efficiency (90% LTV) and protocol sustainability can coexist in decentralized lending. Through automated yield generation delivering 11% APY on LUX collateral, cross-chain integration via M-Chain’s MPC bridge, and conservative risk management, the protocol has processed \$427M in loans since December 2023 without a single liquidation.

Key achievements include:

- 18,400 loans processed with 12,847 active users
- \$31.2M in total yield generated
- Zero liquidations over 2.5 years

- Native Bitcoin support without wrapped tokens
- Integration with Lux’s multi-chain infrastructure

The protocol’s success validates the self-repaying loan model at scale and demonstrates the advantages of cross-chain collateral. Future enhancements including credit lines, synthetic assets, and DAO governance will further strengthen Lux Credit’s position as the most capital-efficient lending protocol in DeFi.

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A Appendix A: Yield Strategy Details

A.1 LUX Staking Strategy

```
class LUXStakingStrategy:
    def __init__(self):
        self.validator_fee = 0.02    # 2% commission
        self.base_reward = 0.142    # 14.2% base APY

    def calculate_apy(self, stake_amount, duration_days):
        # Base rewards
        base_return = stake_amount * self.base_reward * (duration_days / 365)

        # Compound monthly
        monthly_rate = self.base_reward / 12
        compounds = duration_days / 30
        compound_return = stake_amount * ((1 + monthly_rate) ** compounds - 1)

        # Subtract validator fee
        net_return = compound_return * (1 - self.validator_fee)

        return net_return / stake_amount
```

A.2 Liquidity Mining Optimization

Pool Selection Criteria:

1. APY $\geq$ 15% minimum
2. Liquidity $\geq$ \$1M minimum
3. Impermanent loss $\leq$ 5% historical
4. Protocol TVL $\geq$ \$10M
5. Audit status: Verified

B Appendix B: Loan Example

Scenario: Alice deposits 10 BTC when $\text{BTC} = \$60,000$

1. **Collateral Value:** $10 \text{ BTC} \times \$60,000 = \$600,000$
2. **Max Loan (90% LTV):** $\$600,000 \times 0.90 = \$540,000$
3. **Alice borrows:** \$540,000 in luxUSD
4. **BTC Yield:** 4.2% APY = \$25,200/year
5. **Auto-Repayment:** \$25,200/year reduces debt
6. **Loan Duration:** $\$540,000 / \$25,200 = 21.4$ months
7. **Final Position:** Alice keeps all 10 BTC after 21.4 months

Benefit: Alice accessed \$540k liquidity for 21 months while retaining Bitcoin exposure, paying zero interest explicitly.

C Appendix C: MPC Integration Pseudocode

```
# BTC deposit via M-Chain threshold custody
def deposit_btc_collateral(btc_amount, user_address):
    # Generate threshold custody address
    custody_address = mchain.generate_threshold_address(
        threshold=15,
        total_validators=21,
        network="bitcoin"
    )

    # User sends BTC to custody address
    btc_tx = user.send_btc(btc_amount, custody_address)

    # Wait for confirmations
    await btc.wait_for_confirmations(btc_tx, required=6)

    # Verify on M-Chain
    proof = mchain.create_spv_proof(btc_tx)
    verified = mchain.verify_btc_deposit(proof)

    if verified:
        # Calculate max loan at 80% LTV for BTC
        max_loan = btc_amount * btc_price * 0.80
```

```
# Create Lux Credit loan
loan_id = lux_credit.create_loan(
    collateral_asset="BTC",
    collateral_amount=btc_amount,
    loan_amount=max_loan,
    custody_proof=proof
)

return loan_id
```